Transformations

Computer Graphics
Instructor: Sungkil Lee

Today

- **Coordinate systems and frames for affine spaces**
  - Homogeneous coordinates
- **Affine transformations**
  - Rotation, translation, scaling, shear
- **How to build arbitrary transformation matrices**
  - from simple standard transformation matrices

Homogeneous coordinate은 왜 3차원 공간에서
4x4 matrix를 사용하는가?

→ point와 vector를 w값으로 구분하기 위해서

0 : vector

1 : point → 1이 곱해져서 original이 생김

→ translation도 rotation, scaling처럼 4x4 matrix의
곱연산으로 처리하도록 통합하기 위해서

Prerequisites

Linear Independence, Dimension

- **Linear independence**
  - A set of vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n$ is ***linearly independent*** if

$$\alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \cdots + \alpha_n \boldsymbol{v}_n = 0, \qquad \text{iff } \alpha_1 = \alpha_2 = \cdots = 0$$

다른 vector의 조합으로 어느 한 vector를 represent X

  - If a set of vectors are ***linearly independent***, we cannot represent one in terms of the others . Otherwise, if a set of vectors is ***linearly dependent***, at least one can be written in terms of the others

- **Dimension**
  - In a vector space, the maximum number of linearly independent vectors is fixed and is called the ***dimension*** of the space

3차원은 3개 → 축

Basis and Representation

- **Basis** n개의 linear independent vector의 set (다양하게 잡을 수 있음)
  - In an n-dimensional space, any set of n linearly independent vectors form a ***basis*** for the space; such sets are not unique.
  - Given a basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$, any vector $\boldsymbol{v}$ can be written as

$$\boldsymbol{v} = \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \cdots + \alpha_n \boldsymbol{v}_n \,, \quad \text{where the } \{\alpha_i\} \text{ are unique.}$$

Basis가 고정되면 vector의 representation은 unique

- **Representation**
  - The list of scalars $\{\alpha_i\}$ is the representation of v with respect to $\{\boldsymbol{v}_i\}$.
  - We can write the representation as a column vector of scalars.

$$\boldsymbol{a} = [\alpha_1 \alpha_2 \ \ldots \alpha_n]^{\mathrm{T}} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \cdots \\ \alpha_n \end{bmatrix}$$

기준에 따라 같은 object이더라도
representation이 달라지니 transformation 필요

Coordinate-Free Geometry

Coordinate-Free Geometry

- **Coordinate-free geometry**
  - Points exist in space regardless of any reference or coordinate system.
  - Thus, we do not need a coordinate system to specify a point or a vector.
  - Most of geometric results are independent of the coordinate system

point, vector하는 coordinate system이
없어도 존재함

- **Example: Euclidean geometry**
  - Two triangles are identical if two corresponding sides and the angle between them are identical

Coordinate-Free Geometry

- **This fact may seem counter to your experience:**
  - When we learned simple geometry in high school, most of us started with a Cartesian approach; e.g., point p is at locations in space (x, y, z)
  - This approach is nonphysical, because points exist regardless of a coordinate system; as covered in middle-school math.
  - See Figures 3.6 and 3.7 to understand the difference.

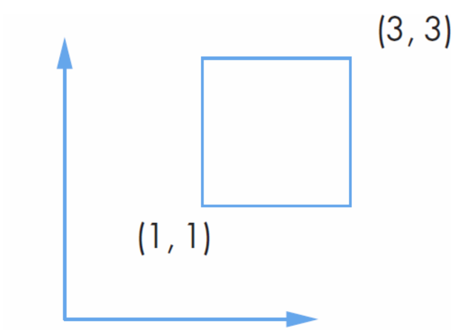


FIGURE 3.6 Object and coordinate system.

X,Y 없이도 object 표현

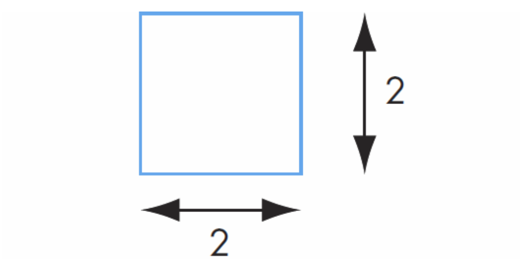


FIGURE 3.7 Object without coordinate system.

Coordinate-Free Geometry

- **We may find it inconvenient to use coordinate-free geometry.**
  - We may need to refer a specific point as "**that blue point over there**."
  - But, it is important to understand that the fundamental geometric relationships are preserved without coordinate systems.
    - The square is still there, or orthogonal lines are still orthogonal, and distances between points remain the same.

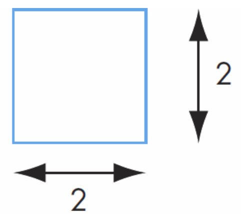


FIGURE 3.7 Object without coordinate system.

  - This reference problem is later solved by coordinate systems and frames.

Coordinate Systems and Frames

Frame of Reference

- **So far, we have been able to work with geometric entities without using any *frame of reference*.**

- **We need a frame of reference to relate points and objects to our physical world.**
  - For example, where is a point?
    - Hard to answer without a reference system
  - Coordinate system examples
    - World coordinates, camera coordinates, screen-space coordinates, …

point와 object에 대한 관계를 갖기 위해서는
reference가 필요함(여러 시스템, 명칭은 다양하지만
동일한 걸 가리킴)

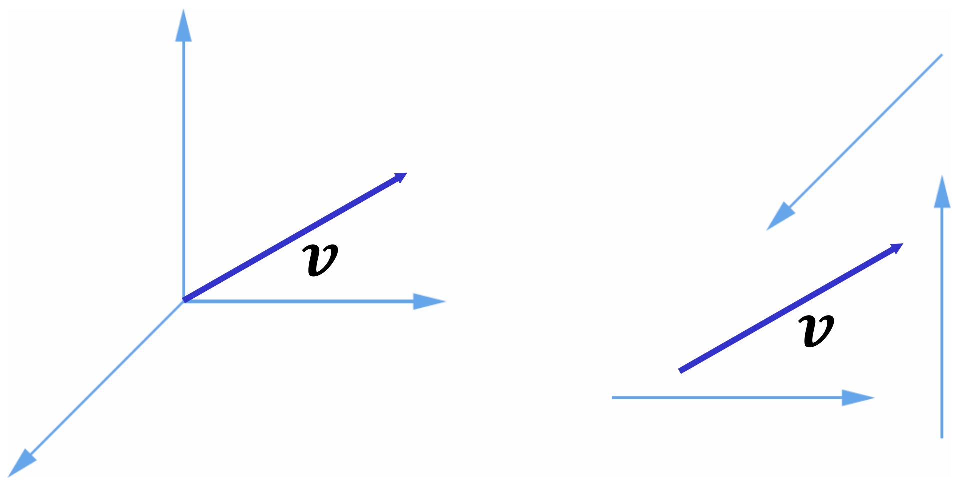
Coordinate Systems

- **An example:**
  - $\boldsymbol{a}$ is a particular representation of a vector $\boldsymbol{v}$, with respect to the basis vectors $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}$ that define a *coordinate system*.

$$\boldsymbol{v} = 2\boldsymbol{v}_1 + 3\boldsymbol{v}_2 - 4\boldsymbol{v}_3$$
$$\boldsymbol{a} = [\,2\;\;3\;-4\,]^{\mathrm{T}}$$

Coordinate Systems

- **Problem**
  - Which is correct?



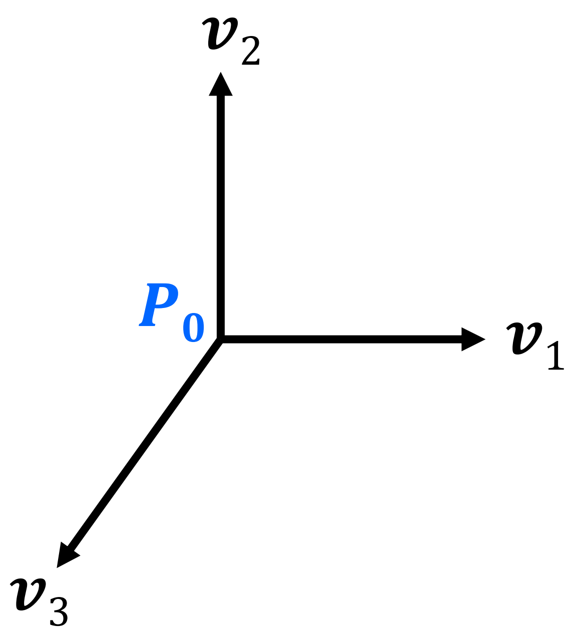
  - Both are correct, because vectors have no fixed locations.

point가 필요 → affine space

Frames

- **A coordinate system is insufficient to represent points,**
  - because points are not defined in a coordinate system.
- **If we work in an *affine space*, we can add a single point, the *origin*, to the basis vectors to form a *frame*.**

Affine space에 오리진을 추가하면 basis vector가 frame



Frames: Representation

- **Suppose a frame determined by** $(P_0, \boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3)$.
  - Within this frame, every vector and point can be written as

$$\boldsymbol{v} = \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \alpha_3 \boldsymbol{v}_3,$$
$$P = P_0 + \beta_1 \boldsymbol{v}_1 + \beta_2 \boldsymbol{v}_2 + \beta_3 \boldsymbol{v}_3$$

origin

  - They appear to have the similar representations, yet a vector has no position.

Frames: Representation

- **If we define** $0 \bullet P$ **=** $\mathbf{0}$ **and** $1 \bullet P = P$ **then we can write them,**

zero vector

$$\boldsymbol{v} = \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \alpha_3 \boldsymbol{v}_3 + \mathbf{0} = [\alpha_1\ \alpha_2\ \alpha_3\ 0][\boldsymbol{v}_1\ \boldsymbol{v}_2\ \boldsymbol{v}_3\ P_0]^{\mathrm{T}}$$
$$P = \beta_1 \boldsymbol{v}_1 + \beta_2 \boldsymbol{v}_2 + \beta_3 \boldsymbol{v}_3 + \boldsymbol{P}_{\mathbf{0}} = [\beta_1\ \beta_2\ \beta_3\ 1][\boldsymbol{v}_1\ \boldsymbol{v}_2\ \boldsymbol{v}_3\ P_0]^{\mathrm{T}}$$

point와 vector에 같은 representation을 사용할 수 있게 됨

  - By this way, we can represent the vectors and points in a single representation.

Homogeneous Coordinates Representations

- **4-D *homogeneous coordinate (HC) representation* represents vectors and points in 3D Cartesian coordinates (CC).**

$$p = [x' \; y' \; z' \; w]^{\mathrm{T}}$$

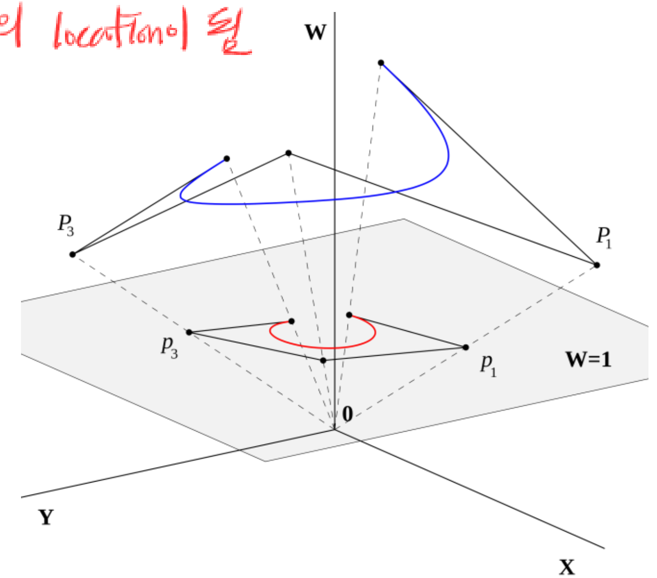
0 = vector

1 = point

  - If $w = 0$, the representation refers to a vector.
  - If $w \neq 0$, the representation refers to a point in 3D.
  - If $w \neq 0$ and $w \neq 1$, we can return to a 3D point by dividing (x', y', z') by w.
    - Perspective division

↳ cartesian coordinate에서의 location이 됨

- **More intuitive example**
  - 2D line in HC = 1D point in CC
  - 1D point in CC is found at w=1 in HC



HCs in Computer Graphics

- **Homogeneous coordinates are keys to all computer graphics systems.**
  - All standard transformations (rotation, translation, and scaling) can be implemented with ***matrix multiplications using 4x4 matrices***.
  - Hence, graphics hardware pipeline is designed to work with ***4-D representations***.

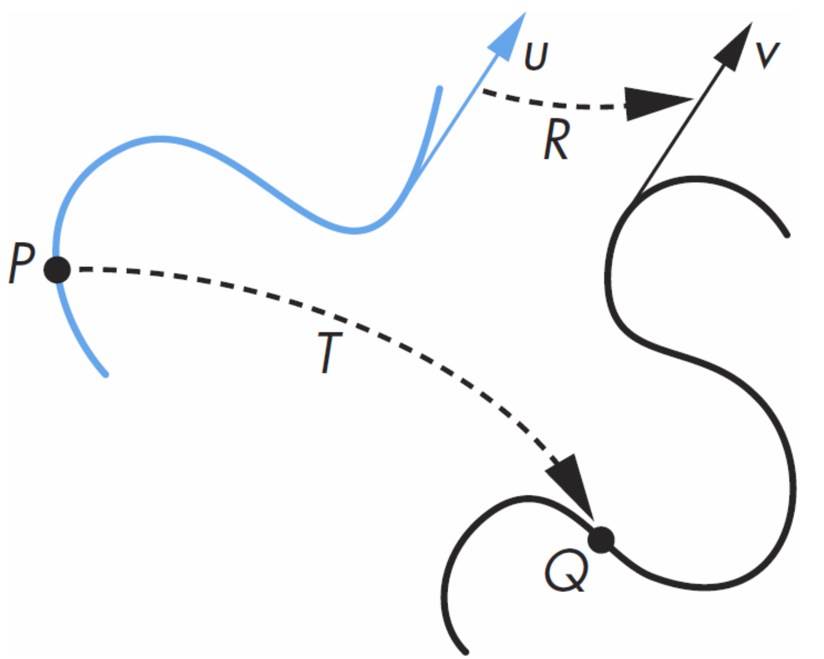
rotation, translation, scaling을
4x4 matrix의 multiplication으로 구현 가능해짐

General vs. Affine Transformations

General Transformations

- **A transformation maps points to other points and/or vectors to other vectors.**

U를 V에 mapping



$$v = R(u)$$

$$Q = T(P)$$

Affine Transformations

- **A matrix for the change of frames is 4×4 and specifies an affine transformation in homogeneous coordinates.**
  - Affine transformation is a function between affine spaces which preserves points, lines, and planes.
  - Every linear transformation is equivalent to a change of frames.

    frame이 변하면 representation도 변하니, representation을 frame에 대해 변환해주는 게 바로 transformation
- **Affine transformations have 12 degrees of freedom (dof).**
  - Upper 4×3 elements of the matrix is defined by 3 dofs from translation, 3 dofs from rotation, 3 dofs from scaling, and 3 dofs from shear transformations.
  - 4 of the elements (the bottom row) in the matrix are fixed.

    ↳ 0001로 고정됨

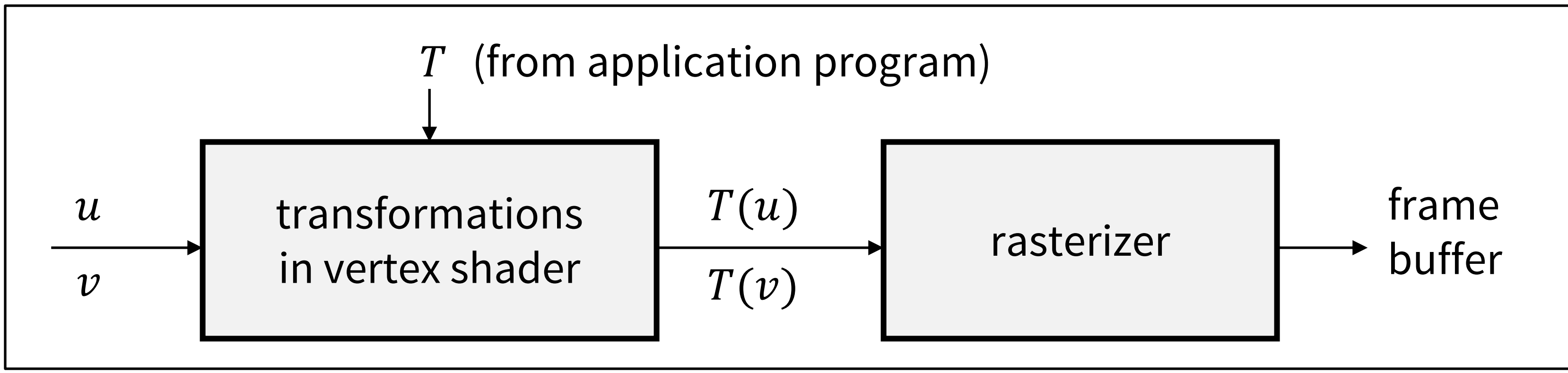
Affine Transformations

- **Affine transformation *preserves lines*.**
  - Lines will remain lines (not become curves or be broken into segments).
  - Characteristic of many physically important transformations
    - Rigid body transformations: translation, rotation → 형태 유지
    - Scaling, shear → 형태를 바꿈

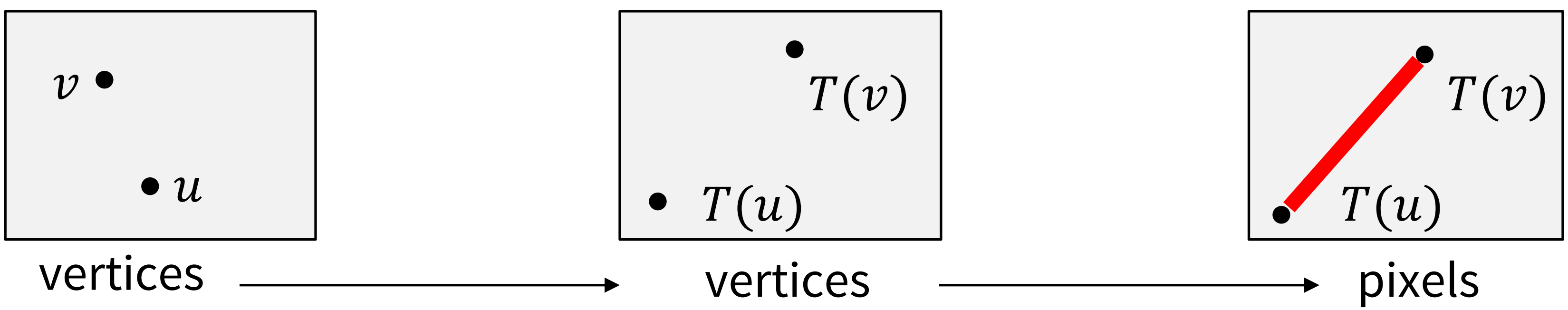
- **Importance in graphics:** line segment의 end point만 transform하면 됨
  - **We can only transform endpoints of line segments** and let implementation draw the line segment by interpolating transformed endpoints.
  - This allows us to realize pipeline approach.
    - Recall that we only transform vertices instead of lines.

Affine Transformations

- **Affine transformation *preserves lines*.**
  - This allows us to realize pipeline approach.



Input, output은 line



두 vertex의 위치를 바꾸고 line으로 연결

Standard (Affine) Transformations

Notation

- We will work with both coordinate-free representations and representations within a HC frame.

P,Q,R: points in an affine space

u,v,w: vectors in an affine space

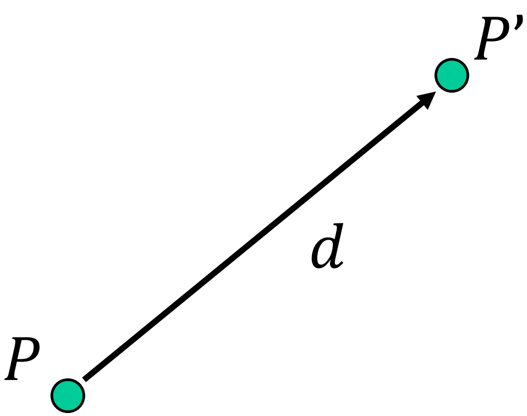
α,β,γ: scalars

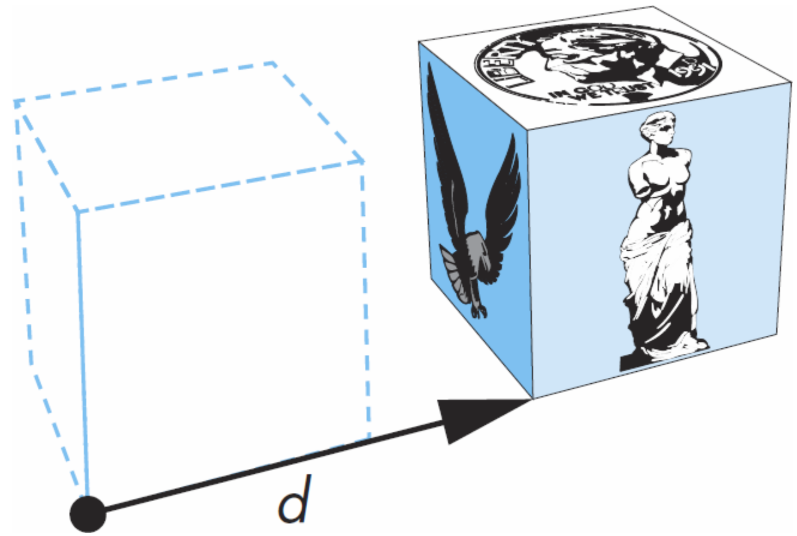
$\mathbf{p},\mathbf{q},\mathbf{r}$: 4-D representations of points in HC

$\mathbf{u},\mathbf{v},\mathbf{w}$: 4-D representations of vectors in HC

Translation 은 point를 new location으로 이동시킴

- **Move (translate, displace) a point to a new location**





- **Displacement determined by a vector** d
  - 3 degrees of freedom
  - $P' = P + d$

Translation Using Representations

- **Using the homogeneous coordinate representation in some frame**

$$\boldsymbol{p} = [\,x\ \ y\ \ z\ \ 1]^{\mathrm{T}}$$
$$\boldsymbol{p}' = [x'\ \ y'\ z'\ \ 1]^{\mathrm{T}}$$
$$\boldsymbol{d} = [d_x\ d_y\ d_z\, 0]^{\mathrm{T}}$$

point + vector = point

- **Hence** $\boldsymbol{p}' = \boldsymbol{p} + \boldsymbol{d}$ **or**

$$x' = x + d_x$$
$$y' = y + d_y$$
$$z' = z + d_z$$

- **Note that this expression is in 4-Ds and expresses point-vector addition.**

Translation Matrix

- **We can also express translation using a 4×4 matrix T in homogeneous coordinates $p' = \mathrm{T}p$ where**

rotation, scaling, shear
모두 포함하려면 translation도
4x4 matrix로 표현해야 함
↓
homogeneous system을
쓰는 이유

Vector

$$\boldsymbol{T} = \boldsymbol{T}(d_x, d_y, d_z) = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$= (x + d_x, y + d_y, z + d_z, 1)$

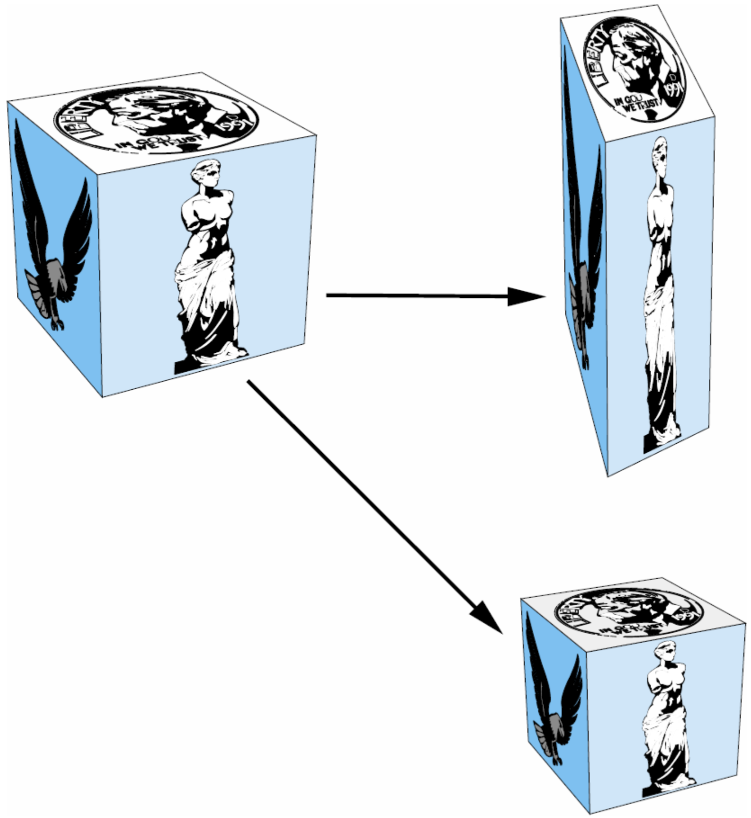
- **This form is better for implementation because**
  - all affine transformations can be expressed in this way (4×4 matrix)
  - and multiple transformations can be concatenated together by multiplying them together.

모든 transform이 multiplying으로 연결됨

Scaling

- **Expand or contract along each axis (fixed point of origin)**

$$x' = s_x x \qquad y' = s_y y \qquad z' = s_z z$$



Scaling Matrix

- **In homogeneous coordinates,**

$$\boldsymbol{p'} = \mathbf{S}\boldsymbol{p}$$

where

translation

$$\mathbf{S} = \mathbf{S}(s_x, s_y, s_z) = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

고정

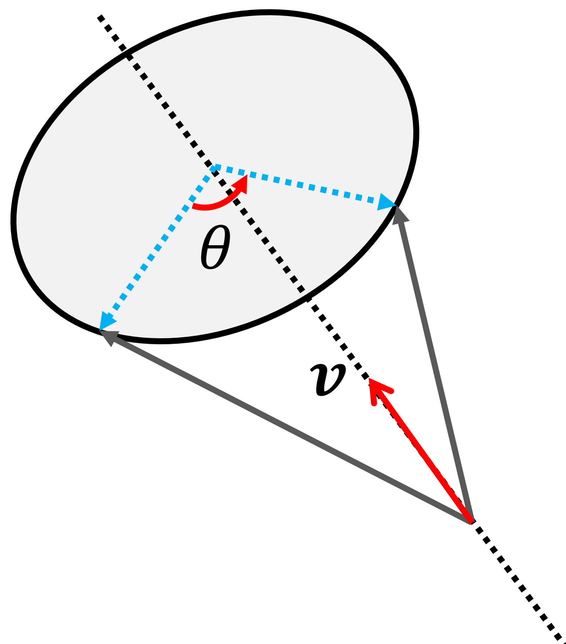
$S_x \cdot x,\ S_y \cdot y,\ S_z \cdot z,\ 1$

Rotation

- **Generally, rotation transformation can be described by the rotation angle θ and its revolution axis v.**

회전축

$$\boldsymbol{p}' = \boldsymbol{R}(\theta)\boldsymbol{p}$$



Rotation Matrix

회전축 v에 대해 θ만큼 돌리는 rotation matrix
(공식 유도가 어려워서 생략)

- **Rotation matrix $R(\theta)$ revolving axis v is given as:**

$v_x=0,\ v_y=0,\ v_z=1$ 이라면

$$\begin{bmatrix} \cancel{v_x}\cancel{v_x}(1-c)+c & \cancel{v_x}\cancel{v_y}(1-c)-\cancel{v_z}s & \cancel{v_x}v_z(1-c)+\cancel{v_y}s & 0 \\ \cancel{v_x}\cancel{v_y}(1-c)+\cancel{v_z}s & \cancel{v_y}\cancel{v_y}(1-c)+c & \cancel{v_y}v_z(1-c)-\cancel{v_x}s & 0 \\ \cancel{v_x}v_z(1-c)-\cancel{v_y}s & \cancel{v_y}v_z(1-c)+\cancel{v_x s} & \cancel{v_z}\cancel{v_z}(1-c)+c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where $c = \cos\theta,\ s = \sin\theta$

$$\begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- This formulation is derived using Quaternion (an extension of complex numbers with three imaginary numbers).
- Though, we do not prove this, because a rigorous proof for this goes far beyond the undergraduate level.

$1-c$

$C=\cos\theta \quad S=\sin\theta$

$x, y, z \cdot S$

$$\begin{array}{c} x \\ y \\ z \end{array} \begin{bmatrix} V_x V_x(1-c)+c & V_x V_y(1-c)-V_z \cdot S & V_x V_z(1-c)+V_y \cdot S \\ V_x V_y(1-c)+V_z \cdot S & V_y V_y(1-c)+c & V_y V_z(1-c)-V_x \cdot S \\ V_x V_z(1-c)-V_y \cdot S & V_y V_z(1-c)+V_x \cdot S & V_z V_z(1-c)+c \end{bmatrix} \begin{array}{c} x \\ x \\ x \end{array}$$

c

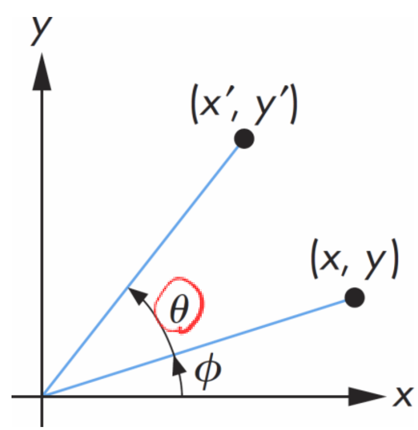
지운 결과
동일해짐

Rotation Matrix

- **Rotation about z axis in 3D**
  - With z-axis ($v_x = 0, v_y = 0, v_z = 1$), the formulation is reduced to the well-known form.
  - equivalent to 2-D rotation in planes of constant z, like slicing 3D into multiple plane slices at height z and rotating in each such plane.

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$



Rotation Matrix

- **Similarly, rotation matrix along x- and y-axes are:**

임의의 축에서도 동작

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

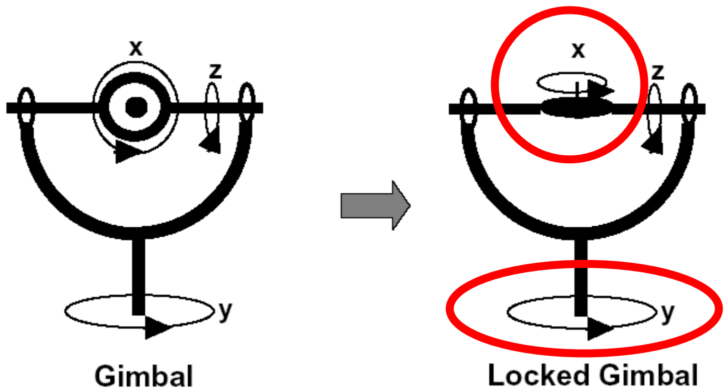
$$\underset{z}{\begin{array}{|c|c|c|} \hline c & -s & \\ \hline s & c & \\ \hline & & 1 \\ \hline \end{array}} \qquad \underset{x}{\begin{array}{|c|c|c|} \hline 1 & & \\ \hline & c & -s \\ \hline & s & c \\ \hline \end{array}} \qquad \underset{y}{\begin{array}{|c|c|c|} \hline c & & s \\ \hline & 1 & \\ \hline -s & & c \\ \hline \end{array}}$$

Rotation Matrix by Euler Angles

2개가 align되는 문제 (gimbal lock)

ㄴ> rotation을 3개로 분리

- **A rotation by θ about an arbitrary axis can be decomposed into the concatenation of rotations about the x, y, and z axes:**
  - $\mathbf{R}(\theta) = \mathbf{R}_x(\theta_x)\mathbf{R}_y(\theta_y)\mathbf{R}_z(\theta_z)$, where $\theta_x, \theta_y, \theta_z$ are called the Euler angles.
  - However, do not apply the rotation matrices with the concatenated form of rotation matrices of Euler angles, because such a rotation may cause a gimbal lock problem in some cases.
  - **Gimbal lock**: when two axes effectively line up, a degree of freedom is lost.



Standard 2D Transformation Matrices

2D Transformation in 4x4 Matrix

- **Use 4x4 matrices instead of 2x2 or 3x3 matrices**
  - Graphics pipeline is optimized for 4x4 matrix, and thus, it is better to use 4x4 matrices even for 2D transformation.
  - It is consistent, when mixing with 3D transformations.

- **It is trivial to derive 2D transformations from 3D transformations.**
  - 2D translation: $\boldsymbol{T} = \boldsymbol{T}(d_x, d_y, 0)$ Z축 고정
  - 2D Scaling: $\mathbf{S} = \mathbf{S}(s_x, s_y, 1)$
  - 2D rotation (with z-axis): $\mathbf{R} = \mathbf{R}_z(\theta)$

General Affine Transformations

Inverse Transformations

여러 transformation을 섞는 경우

- **Basically, we can compute inverse matrices by general formulas.**
  - Though, the inverse operation is very costly, and also, can degrade precision in the computer arithmetic.
- **Alternatively, we can use simple geometric observations:**

$$\mathbf{T}^{-1}(d_x, d_y, d_z) = \mathbf{T}(-d_x, -d_y, -d_z)$$

$$\mathbf{R}^{-1}(\theta) = \mathbf{R}(-\theta) = \mathbf{R}^{\mathrm{T}}(\theta)$$

$$\boldsymbol{S}^{-1}(s_x, s_y, s_z) = \mathbf{S}(1/s_x, 1/s_y, 1/s_z)$$

rotation matrix는 각각의 Basis vector, product의 결과가 0
↳ orthonormal matrix
(normalized된 orthogonal)

General Affine Transformations

- **Multiple transformations can be represented as *concatenation of transformation matrices.***

미리 matrix를 곱해둠

- **We can form arbitrary affine transformation matrices by multiplying rotation, translation, and scaling matrices together.**
  - The cost of forming a matrix $\mathbf{M}=\mathbf{ABCD}$ is insignificant, because the same transformation is applied to many vertices, vertex마다 곱해지는 횟수↓

- **Note that matrix on the right is the first applied:**

$$\boldsymbol{p}' = \mathbf{A}\big(\mathbf{B}(\mathbf{C}\boldsymbol{p})\big) = \mathbf{ABC}\boldsymbol{p}$$

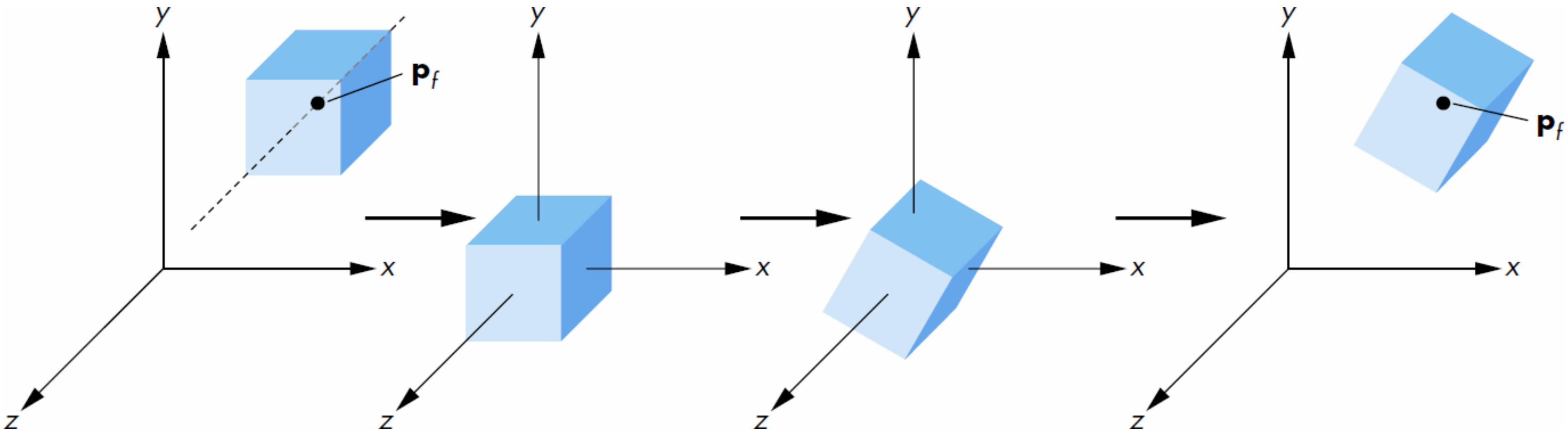
Rotation about Non-Origin Point

1) Move the fixed point to the origin

2) Rotate

3) Move the fixed point back

$$\mathbf{M} = \overset{(3)}{\mathrm{T}(\boldsymbol{p}_f)}\,\overset{(2)}{\mathrm{R}(\theta)}\,\overset{(1)}{\mathrm{T}(-\boldsymbol{p}_f)}$$



Instancing

- **In modeling, we often start with an object centered at the origin, oriented with the axis, and at a standard size.**
  - We apply an ***instance transformation*** to its vertices to scale, orient, and locate somewhere.
  - This allows us to work with minimal geometric objects, while rendering many different objects.